

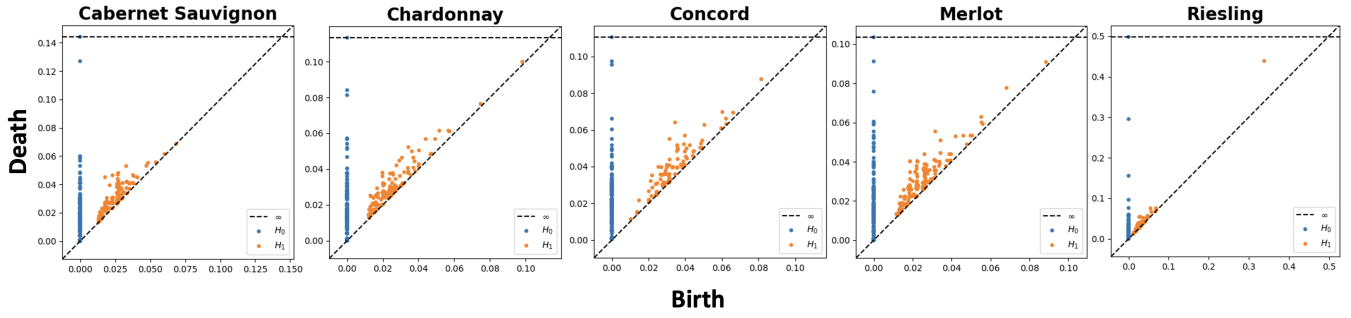


Figure 2: Persistence diagrams for the five grape cultivars, each constructed using their data from seasons 1999 through 2022. Blue and orange dots represent connected components (dgm_0) and holes (dgm_1) respectively. The vertical distance of a dot from the main diagonal corresponds to the feature’s duration.

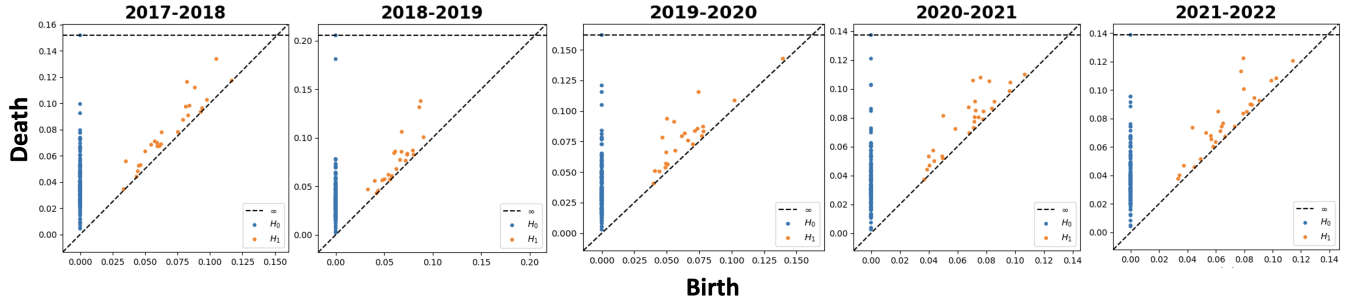


Figure 3: Persistence diagrams for the last 5 seasons of the cold hardiness data. Each diagram is for one season, constructed using all 5 cultivar data for that season.

	CD	CH	CS	MR	WR
CD	0	0.236	0.201	0.305	0.168
CH	0.236	0	0.146	0.164	0.233
CS	0.201	0.146	0	0.168	0.206
MR	0.305	0.164	0.168	0	0.286
WR	0.168	0.233	0.206	0.286	0

Table 2: Pairwise distance matrix for cultivars. For each cultivar data from 1999-2022 was used. Each value shows the Wasserstein distance between the dgm_1 obtained for the corresponding two cultivars.

Results of Intra-cultivar Variability: Next, we ask how variable is each cultivar across the different seasons. This is captured by the level of branching observed within the dgm_1 for that cultivar (Task 1, Section 3.2). Figure 2 shows all five persistence diagrams. As can be readily observed, each cultivar has a different profile. Intuitively, if a cultivar behaves highly variable from season to season, we can expect to see more branching. However, if those branching events are relatively short-lived, then they correspond to small time scale variations. Longer lived branching events correspond to more persistent divergent behavior. From Figure 2, it can be seen that all three of CH, CD, and MR show numerous holes of wide ranging durations. In contrast, CS and WR show fewer holes with smaller duration. This suggests that CS and WR are less variable compared to the other varieties.

Seasonal comparisons: We also compared the different seasons (from 1999 to 2022) using the methodology described in Task 2 of Section 3.2. Figure 3 shows the persistence diagrams for only the last 5 seasons (due to space constraints). We observed that the two most different seasons (i.e., with the largest Wasserstein distance) were the seasons 2001-2002 vs. 2010-2011 (shown in Figure 1).

5 Conclusion

Topological data analysis can be an effective tool to mine for higher order structural information from point cloud data. In this paper, we presented a persistent homology based framework to analyze and glean various types of structural information from a cold hardiness data set. The framework itself is generic and can be extended to other applications within agriculture or other domains. Future research directions include (but are not limited to) a) using the information gained to improve the prediction accuracy of cold hardiness models; b) adverse testing against noise and incomplete data; and c) exploring ways to use relationships inferred toward data imputation and multi-task learning among cultivars.

Acknowledgement

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